

SMART INDIA HACKATHON (SIH) 2026 — COMPLETE TEAM PREPARATION GUIDE

CAREERFIT: AI-POWERED CAREER READINESS & SKILL GAP PLATFORM

Team Size: 6 Members **Internal Hackathon Date:** 9 September 2026 **Location:** College Premises (Offline) **Round:** Ideation / Presentation Round (No Prototype Required)

PART 1 — ABOUT SMART INDIA HACKATHON

1. What is a Hackathon?

A hackathon is a time-bound event where teams come together to solve a specific problem using technology. Think of it as a "problem-solving sprint." You identify a problem, design a solution, build a working prototype, and present it — all within a limited timeframe.

Key characteristics:

- Time-limited (usually 24–48 hours for the final round)
- Problem-focused (you solve a given challenge)
- Team-based (collaboration is essential)
- Presentation-driven (you must explain and demo your solution)
- Innovation-focused (judges look for creative, feasible solutions)

2. What is Smart India Hackathon (SIH)?

Smart India Hackathon is India's largest national-level hackathon organized by the Government of India (Ministry of Education's Innovation Cell — MIC). It brings together students from colleges across India to solve real-world problems faced by various government ministries, departments, and industries.

SIH has two main formats:

- **Software Edition:** Build software solutions (apps, platforms, AI systems, etc.)
- **Hardware Edition:** Build physical/hardware-based solutions

The problems are called **Problem Statements** and they come from real organizations like:

- Ministry of Education
- Ministry of Health
- Ministry of Agriculture
- Various government departments
- Industry partners

3. What Are We Actually Expected to Do?

For the **Internal College Round (Round 1):**

- Select or be given a problem statement
- Understand the problem deeply
- Research users and existing solutions

- Design a proposed solution
- Create an 8–10 minute presentation
- Present before judges
- Answer judge questions (2–3 minutes)

For **Later Rounds (if shortlisted)**:

- Build a working prototype/MVP
- Demonstrate the solution live
- Present to national-level judges
- Defend your technical choices

4. What is a Problem Statement?

A problem statement is a clear, specific description of a real-world problem that needs a technological solution. It defines:

- **What** the problem is
- **Who** faces the problem
- **Why** it needs solving
- **What constraints** exist

Example (hypothetical):

"Students in rural areas lack access to personalized career guidance. There is no affordable, AI-powered platform that analyzes a student's skills and provides a personalized roadmap to become job-ready for specific industry roles."

5. What is a Solution to a Problem Statement?

A solution is your proposed approach to solving the problem. It should:

- Address the specific problem directly
- Be technically feasible with your team's skills
- Provide real value to users
- Be innovative (not just a copy of existing tools)
- Be scalable

6. How is SIH Different from a Normal College Project?

Aspect	College Project	SIH
Problem	Often self-chosen or assigned by teacher	Real-world problem from organizations
Users	Sometimes imaginary	Real, specific users
Judges	College faculty	Government officials, industry experts
Time	Weeks to months	Compressed timeline
Presentation	Often informal	Professional, timed
Innovation	Not always required	Strongly evaluated
Scalability	Rarely considered	Expected to discuss

Feasibility	Sometimes theoretical	Must be practically buildable
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7. What Do Judges Evaluate?

Based on the evaluation criteria:

1. **Innovation & Creativity** — Is this a fresh approach? Does it solve the problem in a unique way?
2. **Technical Feasibility** — Can this actually be built? Is the technology stack appropriate?
3. **Implementation Plan** — Do you have a realistic plan with phases and timeline?
4. **Impact & Scalability** — Will this help real people? Can it grow beyond the prototype?
5. **Presentation & Communication** — Can every member explain the project clearly?

8. What Happens During the Internal College Round?

- Teams present their idea and solution (8–10 minutes)
- Judges ask questions (2–3 minutes)
- Judges evaluate based on the 5 criteria above
- Top teams are shortlisted to represent the college at the national level
- No working prototype is required in Round 1 (ideation round)

9. What Happens If We Are Shortlisted?

- Your team represents the college at the SIH national grand finale
- You typically get more time (24–48 hours) to build or refine a working prototype
- You present before a panel of national-level judges (government + industry)
- Winners receive recognition, funding support, and opportunities for implementation

10. What Happens in Later SIH Stages?

1. **Internal Round (Round 1):** Ideation + presentation (your current stage)
2. **Shortlisting:** Top teams from each college are selected
3. **Grand Finale:** Build demo/prototype + present nationally
4. **Winner Selection:** Based on final presentation, demo, and judge evaluation
5. **Implementation:** Winning solutions may be adopted by the problem statement organization

11. Why is Problem Selection Extremely Important?

- If you choose a problem you don't understand, you cannot build a good solution
- If you force your existing project into an unrelated problem, judges will see through it
- The problem defines everything: users, solution, AI, technology, impact
- A strong problem with a moderate solution beats a weak problem with a fancy solution
- Judges first evaluate if the problem is real and significant

The right approach:

```

PROBLEM FIRST
→ UNDERSTAND USERS
→ DESIGN SOLUTION
→ DESIGN AI
→ DESIGN ARCHITECTURE
→ SELECT TECHNOLOGY
→ PLAN IMPLEMENTATION

```

- PRESENT
- BUILD AFTER SHORTLISTING

The wrong approach:

- CAREERFIT
- FIND A WAY TO FORCE IT INTO SIH

12. Why Should We Understand the Problem Before Writing Code?

- Code without understanding is wasted effort
 - You may build the wrong thing entirely
 - Judges will ask "why did you make this choice?" — you need real answers
 - Understanding the problem leads to better architecture decisions
 - Understanding users leads to better UX decisions
 - Understanding constraints leads to realistic implementation plans
-

PART 2 — OUR CURRENT PROJECT: CAREERFIT

What CareerFit Currently Does

CareerFit is an AI/career-oriented platform that helps students and job seekers understand how well their current skills match industry/job requirements.

Core Concept:

```
RESUME + JOB DESCRIPTION
    ↓
SKILL EXTRACTION
    ↓
MATCHED SKILLS
    ↓
MISSING SKILLS
    ↓
CAREER READINESS / MATCH SCORE
    ↓
SKILL GAP ANALYSIS
    ↓
RECOMMENDATIONS
    ↓
PERSONALIZED LEARNING ROADMAP
```

Existing Capabilities:

- Resume analysis (PDF/DOCX upload)
- Job description analysis
- Skill extraction from text
- Matching resume skills with job requirements
- Identifying missing skills
- Basic match score

- Weighted match score
- Skill-gap analysis
- Recommendations
- End-to-end analysis pipeline
- Backend APIs
- Testing suite

Example:

Resume Skills: Python, Git, GitHub, REST API

Job Requirements: Python, Git, GitHub, REST API, SQL, FastAPI, Docker, Linux

Matched: Python, Git, GitHub, REST API

Missing: SQL, FastAPI, Docker, Linux

Readiness Score: 50% (4/8 skills matched)

Important Note:

CareerFit is a **TECHNICAL FOUNDATION**, not a finished SIH product. If it matches an appropriate SIH problem statement, we propose an **enhanced AI-powered career-readiness and skill-gap solution**, and our existing work gives us a technical head start.

Do NOT present CareerFit as:

"We already built a website and want to submit it."

Do present it as:

"We are proposing an enhanced AI-powered career-readiness and skill-gap solution. Our existing work gives us a technical foundation."

PART 3 — CAREERFIT'S SIH THEME

Why CareerFit Belongs to Smart Education

CareerFit fits naturally under the **Smart Education** theme because it addresses:

Smart Education Theme	CareerFit Connection
Student career readiness	Core focus of CareerFit
Employability	Helps students become job-ready
Skill-gap identification	Primary feature
Industry requirements	Maps student skills to industry needs
Personalized learning	Provides customized learning roadmaps
Career guidance	Recommends career paths
Education-to-employment transition	Bridges the gap between academics and industry

Stronger Positioning

Weak positioning:

"Resume-job matching website"

Strong positioning:

"An AI-powered career-readiness and skill-gap platform that helps students understand the gap between their current capabilities and industry requirements and provides a personalized pathway to become job-ready."

When CareerFit Should NOT Be Used

If the available SIH problem statement does not genuinely match CareerFit, we should **NOT** force CareerFit into it. The rule is:

OFFICIAL PROBLEM

→ REAL USER NEED

→ CAREERFIT FIT

→ ENHANCEMENT

NOT:

CAREERFIT

→ FIND A WAY TO FORCE IT INTO SIH

If CareerFit doesn't fit a specific problem statement, the team should explore other solutions that genuinely address the problem.

PART 4 — THE BIGER CAREERFIT VISION

AI-Powered Career Readiness and Personalized Skill Development Platform

Full Vision — 20 Potential Capabilities:

#	Capability
1	Student profile
2	Resume upload
3	Skill extraction
4	Job-description analysis
5	Industry skill-demand analysis
6	Skill matching
7	Missing-skill detection

8	Skill importance/priority
9	Career readiness score
10	Explainable scoring
11	Personalized learning roadmap
12	Recommended courses/resources
13	Career-path recommendations
14	Progress tracking
15	Skill improvement tracking
16	Resume improvement suggestions
17	Job-role recommendations
18	Industry trend analysis
19	Personalized recommendations
20	AI career assistant

Feature Prioritization:

MUST HAVE (MVP)

1. Student profile
2. Resume upload and parsing
3. Skill extraction (NLP-based)
4. Job-description analysis
5. Skill matching
6. Missing-skill detection
7. Weighted skill priority
8. Career readiness score
9. Explainable scoring
10. Personalized learning roadmap

SHOULD HAVE (V2)

11. Recommended courses/resources
12. Career-path recommendations
13. Resume improvement suggestions
14. Job-role recommendations
15. Industry trend analysis

COULD HAVE (V3)

16. Progress tracking
17. Skill improvement tracking
18. Institution dashboard
19. Placement analytics

FUTURE

- 20. AI career assistant
- 21. Multilingual support
- 22. Industry partnerships
- 23. Skill trend prediction
- 24. Learning platform integration

The MVP remains achievable — focus on 10 core features.

PART 5 — IMPORTANT INNOVATION DIRECTION

Beyond Simple Resume Comparison

CareerFit should not simply compare a resume and job description. It should answer:

"How job-ready is this student, WHY are they at that level, WHAT skills are missing, WHICH skills matter most, and WHAT should they do next?"

Enhanced Workflow:



Why This Is More Than a Recruitment Website

Recruitment Website	CareerFit
Matches candidates to jobs	Analyzes skill readiness

Focuses on hiring	Focuses on learning
Binary: hired/not hired	Spectrum: readiness score
No explanation	Full explanation
No learning path	Personalized roadmap
Employer-centric	Student-centric
One-time assessment	Continuous improvement

CareerFit is **educational**, not just functional. It teaches students what to learn next, not just whether they qualify.

PART 6 — AI INSIDE CAREERFIT

Two Different Uses of AI

A. AI Used to BUILD the Project (Development Tools)

These are tools that help you code and develop faster:

Tool	Purpose
Claude / ChatGPT	Research, documentation, brainstorming
Cursor / Claude Code	Code generation, debugging
Other assistants	Test generation, UI design

Use these to help with:

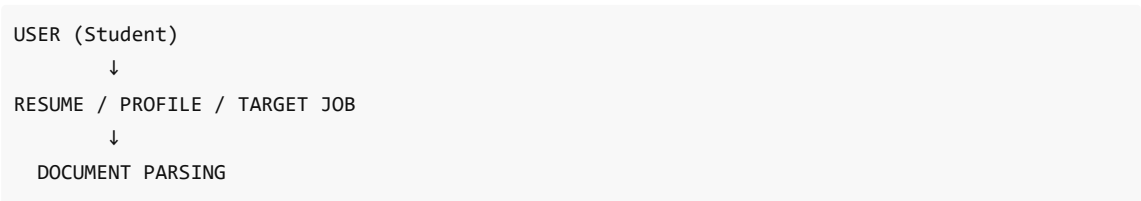
- Coding and debugging
- Research and learning
- UI design ideas
- Documentation
- Architecture brainstorming
- Test generation
- Presentation preparation

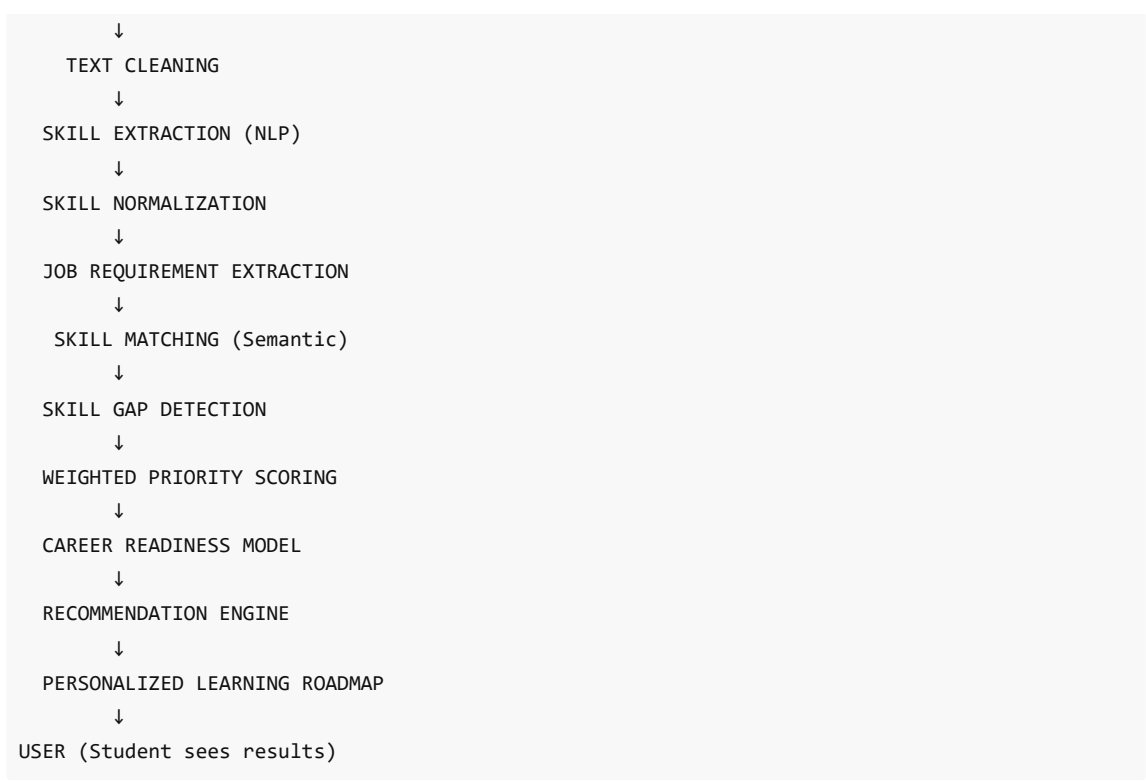
These are development aids, not part of the product.

B. AI Inside the Actual CareerFit Product (What Matters)

This is the AI that runs inside the platform and provides value to users.

AI Workflow Design:





Potential AI Technologies:

Technology	Purpose	Why Needed
NLP	Extract skills from text	Resume and job descriptions are text
Embeddings	Semantic understanding	"ML" and "Machine Learning" mean the same thing
Semantic Similarity	Match skills accurately	Skills have synonyms and variations
LLMs	Generate explanations and roadmaps	Natural language output
ML Models	Score readiness	Pattern-based scoring
Recommendation Systems	Suggest courses/paths	Personalized suggestions

Critical: Every AI Component Must Have a Purpose

For each AI component, answer:

Question	Answer Required
Why is the AI needed?	What problem does it solve?
What data enters the AI?	Input specification
What does the AI produce?	Output specification

How is the result validated?	Accuracy checking
How are errors handled?	Fallback mechanisms
How is accuracy evaluated?	Metrics and testing
What if the AI API fails?	Graceful degradation

Example — Skill Extraction:

Aspect	Detail
Why needed	Skills are buried in free-form resume text
Input	Resume text (PDF/DOCX content)
Output	List of extracted skills
Validation	Compare against known skill databases
Error handling	If extraction fails, allow manual input
Accuracy	Test against 100+ resumes, measure precision/recall
API failure	Fall back to keyword-based extraction

PART 7 — EXPLAINABILITY

Why "Your Score is 68%" Is Not Enough

A bare score tells the student nothing actionable. They don't know:

- Why it's 68% and not 85%
- Which skills are helping their score
- Which skills are hurting their score
- What to do to improve

The Explainable Output:

```
CAREER READINESS SCORE: 68%

WHY THIS SCORE:
├─ STRONG COVERAGE (positive impact):
|   ├── Python – Strong match
|   ├── Git – Strong match
|   └── REST APIs – Strong match
|
├─ CRITICAL GAPS (negative impact):
|   ├── SQL – Required for 85% of data roles
|   ├── FastAPI – Required for backend development
|   └── Docker – Required for deployment
|
```

RECOMMENDATION:
Priority: SQL → FastAPI → Docker
Suggested courses and resources provided.

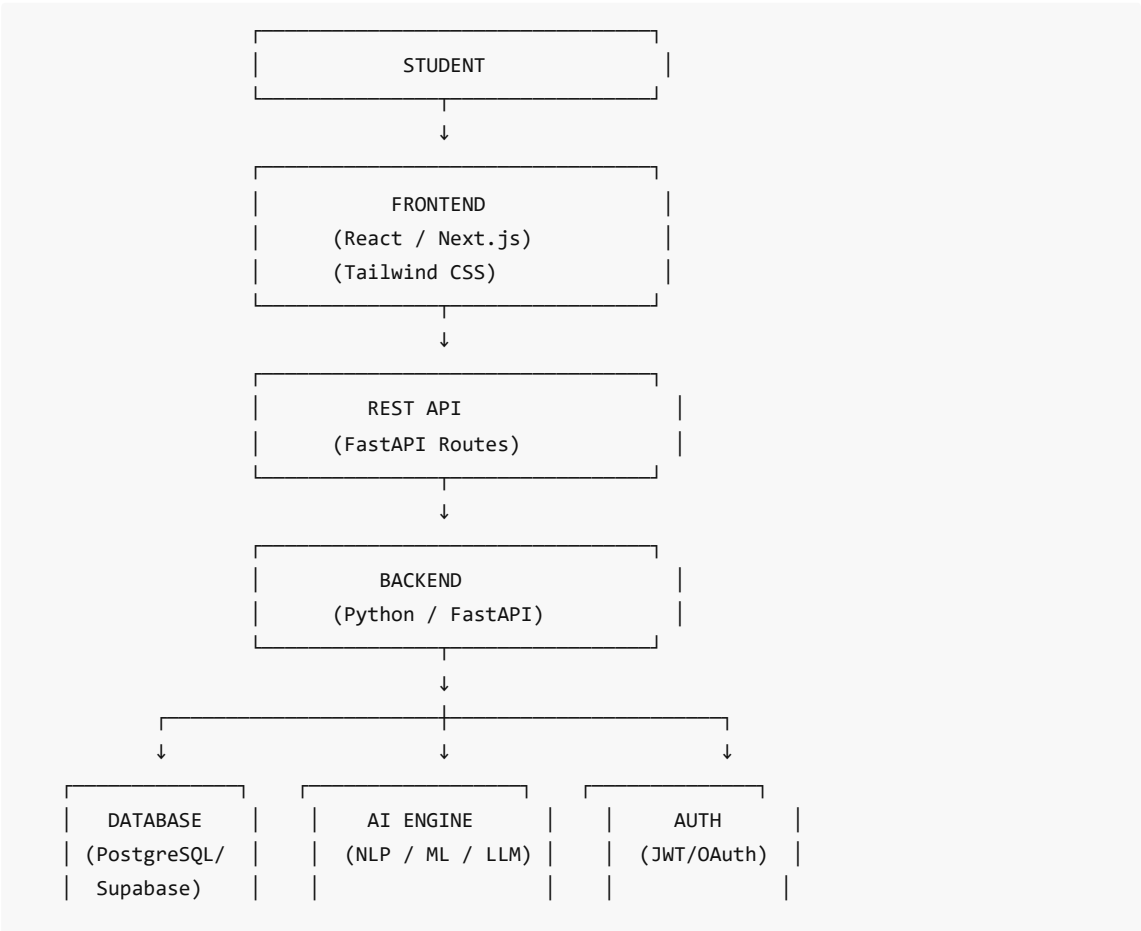
Why Explainability Matters

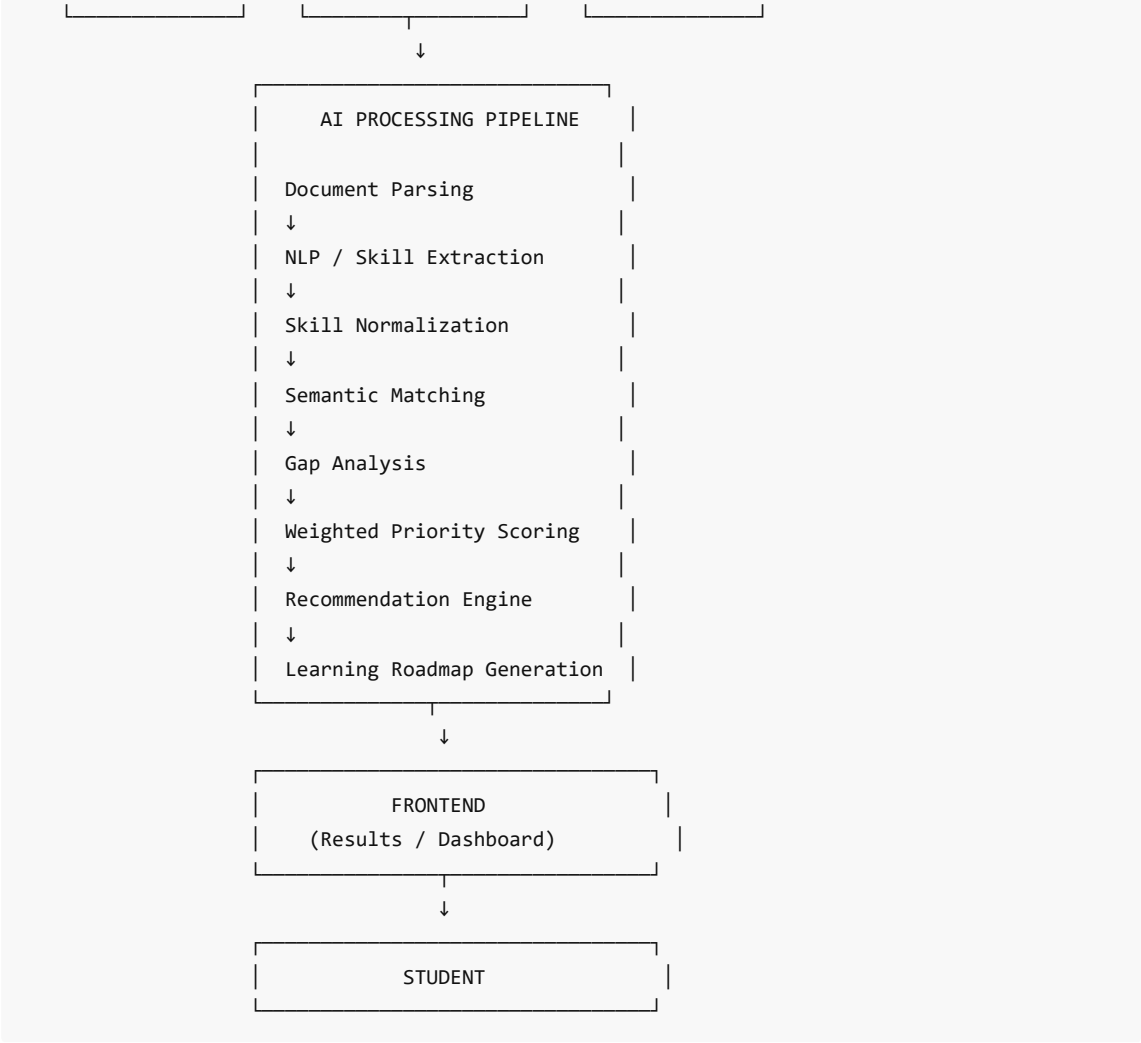
Without Explanation	With Explanation
"Your score is 68%"	"Your score is 68% because you have Python and Git, but lack SQL and Docker"
Student feels confused	Student understands exactly where they stand
No clear next step	Clear priority: Learn SQL first, then FastAPI, then Docker
Low trust	Higher trust — the system shows its reasoning
One-time interaction	Students return to track improvement

Explainability transforms CareerFit from a "score generator" into a "career coach."

PART 8 — POSSIBLE CAREERFIT ARCHITECTURE

High-Level Architecture





Technology Stack with Justifications

Frontend

Technology	Why Selected
React / Next.js	Component-based, fast rendering, SEO-friendly, large ecosystem
Tailwind CSS	Rapid UI development, consistent design, no custom CSS overhead

Backend

Technology	Why Selected
Python	Best language for AI/ML, rich ecosystem, fast development
FastAPI	High performance, automatic API docs, async support, type-safe

Database

Technology	Why Selected
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PostgreSQL	Reliable, handles structured data, ACID compliant
Supabase	PostgreSQL as a service, built-in auth, easy setup

AI/ML

Technology	Why Selected
NLP Libraries (spaCy, NLTK)	Text processing, skill extraction
Embeddings (sentence-transformers)	Semantic understanding of skills
LLM API (OpenAI/Claude)	Natural language explanations, roadmap generation
ML Models (scikit-learn)	Scoring, classification, pattern recognition

File Processing

Technology	Why Selected
PyPDF2 / pdfplumber	PDF parsing
python-docx	DOCX parsing

Deployment

Technology	Why Selected
Vercel (frontend)	Free tier, fast deployment, CI/CD
Cloud (backend)	Scalable hosting for Python backend

Version Control

Technology	Why Selected
Git	Industry standard, version tracking
GitHub	Collaboration, code review, CI/CD

Rule: Don't add technologies just to look impressive. Every technology must solve a specific problem.

PART 9 — SIX-MEMBER TEAM

MEMORBER 1 — TEAM LEADER / PRODUCT & PROBLEM LEAD

Responsibilities:

- Understand SIH completely
- Understand the official problem statement deeply
- Research users and their needs

- Define and justify the problem
- Coordinate all team members
- Make final decisions on direction
- Lead overall presentation
- Understand the entire project end-to-end

Presentation Section: Problem & Introduction

Must be able to answer:

- What is the problem?
 - Who has the problem?
 - Why is it important?
 - Why did we choose it?
 - What is our project in one sentence?
-

MEMBER 2 — SOLUTION / UX LEAD

Responsibilities:

- Design user journey and experience
- Define major features
- Create wireframes and user flows
- Design the proposed solution workflow
- Explain how CareerFit solves the problem
- Ensure the solution is user-friendly

Presentation Section: Proposed Solution + User Journey

Must explain:

Student → Upload/Input → Analysis → Skill Gap → Recommendations → Roadmap → Progress

MEMBER 3 — AI/ML LEAD

Responsibilities:

- Design AI architecture
- NLP pipeline for skill extraction
- Semantic matching implementation
- Embeddings and similarity logic
- Recommendation logic
- AI evaluation and testing
- Handle AI limitations and edge cases
- Explainability implementation
- AI safety and validation

Presentation Section: AI Intelligence Workflow

Must know:

- Where AI is used in the system
- Why AI is needed at each point
- What goes INTO the AI

- What comes OUT of the AI
 - How AI results are validated
 - How hallucinations/errors are controlled
-

MEMBER 4 — BACKEND / SYSTEM ARCHITECT

Responsibilities:

- Backend API development
- Database design and management
- Authentication and security
- Data flow between components
- System architecture
- Integration between frontend and AI engine
- API documentation

Presentation Section: System Architecture

Must understand:

Frontend → API → Backend → Database → AI → Recommendation → Frontend

MEMBER 5 — DEVELOPMENT / TECHNOLOGY LEAD

Responsibilities:

- Technology stack decisions
- Development planning and phases
- Git/GitHub workflow
- Code integration
- Testing strategy
- Deployment pipeline
- Timeline management
- MVP planning

Presentation Section: Technology Stack + Implementation Plan

Must explain:

- Why each technology was chosen
 - What gets built first (MVP)
 - What comes later
 - How long each phase takes
 - How testing works
-

MEMBER 6 — IMPACT / SCALABILITY / PRESENTATION LEAD

Responsibilities:

- Expected impact analysis
- Education impact assessment
- Employability impact measurement

- Scalability planning
- Future scope definition
- Presentation conclusion
- Coordinate judge Q&A responses

Presentation Section: Impact + Scalability + Conclusion

Must explain:

- Who benefits from this solution?
- What changes for the better?
- How do we measure success?
- Can this scale to more users?
- What can be added in the future?

PART 10 — EVERY MEMBER MUST KNOW THESE

Even though each person has a specific role, **EVERY member** must know:

#	Topic
1	Problem statement and justification
2	Target users
3	Proposed solution
4	AI workflow
5	System architecture
6	Innovation and differentiation
7	Technology stack
8	Implementation plan
9	Expected impact
10	Limitations
11	Future scope

Nobody should say: "That is not my part."

A judge can ask **any member anything**. If Member 6 cannot explain the AI workflow, it creates a bad impression.

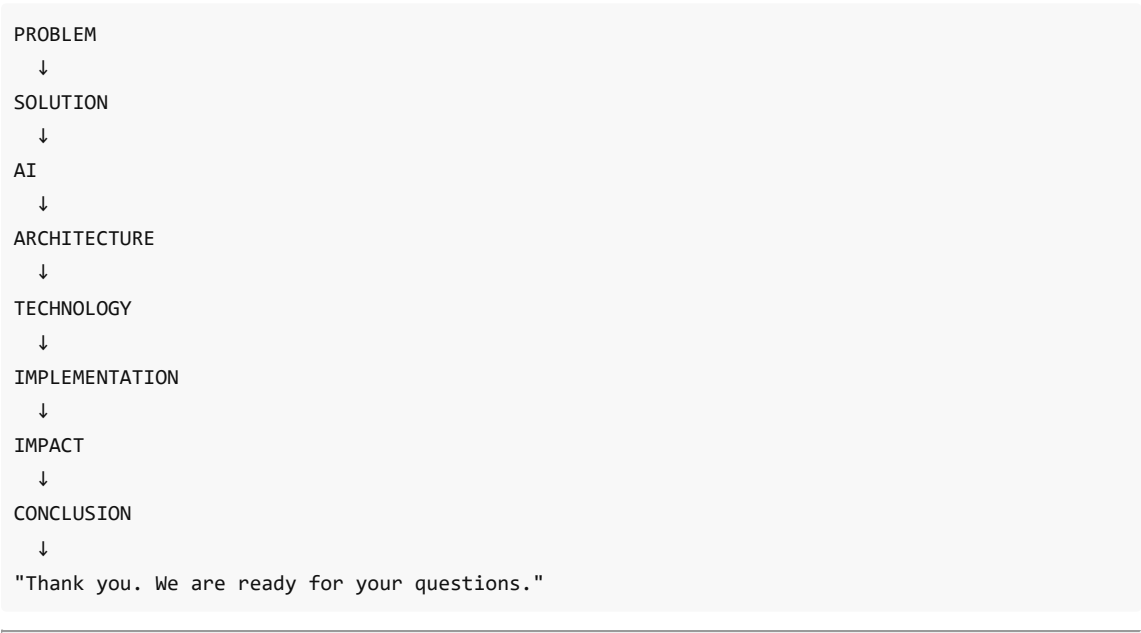
PART 11 — 8–10 MINUTE PRESENTATION STRUCTURE

8 Slides — Exact Structure

Slide	Topic	Presenter	Time
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1	Problem Statement & Justification	Member 1	1:30
2	Proposed Solution	Member 2	1:30
3	AI / Intelligence Workflow	Member 3	1:30
4	System Architecture	Member 4	1:00
5	Technology Stack	Member 5	0:45
6	Implementation Plan & Timeline	Member 5	0:45
7	Expected Impact & Scalability	Member 6	1:00
8	Conclusion / Why Our Solution Matters	Member 6	0:30
—	Buffer	—	0:30
—	TOTAL	—	~9:00

Presentation Flow:



PART 12 — SIX PEOPLE'S PRESENTATION TOPICS

Member	Topic	Duration
Member 1	Problem & Why It Matters	~1:30
Member 2	Our Proposed Solution	~1:30
Member 3	AI Intelligence	~1:30
Member 4	System Architecture	~1:00

Member 5	Technology Stack & Implementation	~1:30
Member 6	Impact, Scalability & Conclusion	~1:30

PART 13 — SLIDE 1 SCRIPT (MEMBER 1)

"Problem & Why It Matters"

[WALK TO FRONT]

"Good morning respected judges.

We are Team [TEAM NAME], and today we are addressing a problem that affects millions of students across India."

[PAUSE — LOOK AT JUDGES]

"Students complete degrees, earn certifications, and build resumes — but most of them don't actually know whether their skills match what the industry requires."

[POINT TO SLIDE]

"Let me give you a real scenario. A student learns Python, Git, and builds a few projects. They apply for a backend developer role. But the job requires SQL, Docker, FastAPI, and Linux — skills they've never touched. They get rejected, and they don't even know why."

[PAUSE]

"This is not a hypothetical problem. Studies show that a significant percentage of Indian graduates are considered 'unemployable' — not because they lack intelligence, but because there is a gap between what they learned and what the industry expects."

[GESTURE]

"Today, there is no simple, affordable, AI-powered tool that tells a student: 'Here is exactly where you stand, here is what's missing, and here is what you should learn next — in order of importance.'"

[LOOK AT JUDGES]

"That is the problem we are solving.

Our question is simple: **How job-ready is this student, and what should they do to become job-ready?**

That is what our project addresses. Thank you."

[STEP BACK — TRANSITION TO MEMBER 2]

Time: ~1:30 minutes

Transition to Member 2: "Now, let me hand over to [Member 2's name], who will walk you through our proposed solution."

PART 14 — SLIDE 2 SCRIPT (MEMBER 2)

"Our Proposed Solution"

[STEP FORWARD]

"Thank you, [Member 1].

We propose CareerFit — an AI-powered career-readiness and skill-gap platform."

[POINT TO SLIDE — USER JOURNEY]

"Let me walk you through how a student would use CareerFit."

[TRACE THE FLOW]

"Step 1: A student creates a profile and uploads their resume.

Step 2: The system uses AI to extract skills from the resume — technical skills, soft skills, tools, frameworks — everything.

Step 3: The student selects or inputs their target job role — for example, 'Backend Developer' or 'Data Analyst.'

Step 4: CareerFit analyzes the job description and extracts the required skills for that role.

Step 5: Now comes the key part — the system compares the student's current skills against the job requirements and identifies the gaps."

[PAUSE]

"But we don't stop at listing missing skills. We answer three critical questions:

One: What skills are you missing? **Two:** Which of those skills matter most for your target role? **Three:** What should you learn, in what order, to become job-ready?"

[POINT TO ROADMAP SECTION]

"Step 6: The student receives an explainable career-readiness score — not just a number, but a clear explanation of why they got that score.

Step 7: Finally, they get a personalized learning roadmap — a prioritized path of courses, resources, and skills to learn."

[LOOK AT JUDGES]

"This is not a resume checker. This is a career coach powered by AI that tells students exactly where they stand and exactly what to do next."

[STEP BACK]

Time: ~1:30 minutes

Transition to Member 3: "To explain how the AI actually works behind the scenes, here is [Member 3's name]."

PART 15 — SLIDE 3 SCRIPT (MEMBER 3)

"AI Intelligence Workflow"

[STEP FORWARD]

"Thank you, [Member 2].

I will explain how AI powers CareerFit. Let me walk you through the complete AI workflow."

[POINT TO DIAGRAM]

"It starts with the student's resume and target job description.

Step 1: Document Parsing. The system reads the uploaded PDF or DOCX file and extracts the raw text. We use Python libraries — PyPDF2 and python-docx — for this."

[PAUSE]

Step 2: Text Cleaning. The raw text goes through cleaning — removing special characters, normalizing whitespace, and preparing it for analysis."

[MOVE TO NEXT STEP]

Step 3: Skill Extraction — this is where NLP comes in. We use Natural Language Processing to identify skills mentioned in the resume. Not just exact keyword matching — semantic understanding. For example, if a student writes 'ML' or 'machine learning' or 'statistical modeling,' the system should understand these are related skills."

[PAUSE]

Step 4: Skill Normalization. Skills have many variations. 'JS' and 'JavaScript' are the same. 'React.js' and 'React' are the same. The system normalizes all skill variations to a standard form."

[GESTURE]

Step 5: Semantic Matching. This is a key AI component. We use embeddings — numerical representations of skills — to understand semantic similarity. When a job says 'experience with relational databases,' the system should connect that to SQL, PostgreSQL, MySQL."

[PAUSE]

Step 6: Skill Gap Detection. After matching, the system identifies exactly which skills the student has and which are missing."

[POINT TO PRIORITY]

Step 7: Weighted Priority. Not all missing skills are equally important. If a job requires SQL and Docker, and SQL appears in 90% of similar job postings while Docker appears in 50%, SQL gets higher priority."

[PAUSE]

Step 8: Career Readiness Score. The system calculates a score based on weighted coverage — how many important skills does the student have?"

[POINT TO EXPLANATION]

Step 9: Explainability. We don't just show a number. We explain WHY the student got that score — which skills helped, which are missing, and why it matters."

[GESTURE]

Step 10: Personalized Learning Roadmap. Finally, using an LLM, we generate a clear, prioritized learning path — what to learn first, second, and third — with suggested resources."

[LOOK AT JUDGES]

"Every AI component has a clear purpose, a defined input and output, and validation mechanisms. AI is not used for decoration — it is used because the problem requires understanding natural language, matching semantics, and generating personalized guidance."

[STEP BACK]

Time: ~1:30 minutes

Transition to Member 4: "To explain the technical system architecture, here is [Member 4's name]."

PART 16 — SLIDE 4 SCRIPT (MEMBER 4)

"System Architecture"

[STEP FORWARD]

"Thank you, [Member 3]."

Let me show you how all the components of CareerFit work together."

[POINT TO ARCHITECTURE DIAGRAM]

"At the top, we have the **Student** interacting with the **Frontend** — a clean, responsive web interface built with React and Next.js."

[TRACE THE FLOW]

"When the student uploads a resume or requests analysis, the Frontend sends a request to our **Backend API** — built with FastAPI in Python."

[PAUSE]

"The Backend handles three key responsibilities:

First: Authentication — Students sign up, log in, and their data is protected. We use JWT-based authentication.

Second: Data Management — Student profiles, resume data, and analysis history are stored in a **PostgreSQL database** (via Supabase).

Third: AI Engine Integration — The Backend sends the resume and job description to the AI Engine for processing."

[POINT TO AI SECTION]

"The **AI Engine** is the core processing unit. It runs the full pipeline: document parsing → skill extraction → normalization → matching → gap analysis → scoring → recommendations."

[PAUSE]

"Once the AI Engine completes processing, results flow back through the Backend API to the Frontend — where the student sees their matched skills, missing skills, readiness score, explanation, and learning roadmap."

[LOOK AT JUDGES]

"The architecture is designed to be modular. Each component — Frontend, Backend, Database, AI Engine — can be developed, tested, and improved independently. This makes our system maintainable and scalable."

[PAUSE]

"For deployment, the Frontend goes on Vercel, and the Backend runs on a cloud platform. We use Git and GitHub for version control throughout development."

[STEP BACK]

Time: ~1:00 minute

Transition to Member 5: "To explain our technology choices and implementation plan, here is [Member 5's name]."

PART 17 — SLIDES 5 AND 6 SCRIPT (MEMBER 5)

"Technology Stack & Implementation Plan"

[STEP FORWARD]

"Thank you, [Member 4].

Let me explain why we chose each technology and how we plan to build CareerFit."

TECHNOLOGY STACK

[POINT TO SLIDE 5]

"**Frontend: React / Next.js with Tailwind CSS.** Why? React gives us a component-based architecture — reusable, maintainable UI pieces. Next.js adds server-side rendering and faster page loads. Tailwind CSS lets us build a clean, professional interface quickly."

[PAUSE]

"**Backend: Python with FastAPI.** Why? Python is the standard language for AI/ML. FastAPI is modern, fast, and automatically generates API documentation. It handles async requests efficiently — important when AI processing takes time."

[PAUSE]

"**Database: PostgreSQL via Supabase.** Why? PostgreSQL is reliable and handles structured data well. Supabase gives us PostgreSQL as a service with built-in authentication — less infrastructure to manage."

[PAUSE]

"**AI/ML: NLP + Embeddings + LLM.** Why? NLP extracts skills from text. Embeddings enable semantic matching. LLMs generate natural language explanations and roadmaps. Each serves a specific purpose."

[PAUSE]

"**File Processing: PyPDF2 and python-docx.** Why? Students upload resumes in PDF and DOCX formats. These libraries handle parsing reliably."

[PAUSE]

"**Version Control: Git and GitHub.** Why? Industry standard. All six team members can collaborate, review code, and track changes."

[TRANSITION TO SLIDE 6]

IMPLEMENTATION PLAN

[POINT TO TIMELINE]

"We have planned seven phases:"

"**Phase 1: Problem Validation & Research** — Understand the problem statement, research users, study existing solutions. This is where we are now."

"**Phase 2: UI Design + Backend Setup** — Build the frontend interface and set up the backend with database and authentication."

"**Phase 3: Resume & Job Parsing** — Implement PDF/DOCX parsing and basic text extraction."

"**Phase 4: AI Skill Extraction & Matching** — Build the NLP pipeline, skill extraction, normalization, and semantic matching."

"**Phase 5: Recommendation Engine** — Implement scoring, gap analysis, explainability, and learning roadmap generation."

"**Phase 6: Testing & Refinement** — Test with real resumes, validate accuracy, fix edge cases, refine the UI."

"**Phase 7: Deployment & Presentation Prep** — Deploy the working prototype and prepare for the final presentation."

[LOOK AT JUDGES]

"Each phase builds on the previous one. We start with the foundation and layer intelligence on top. This ensures we always have something functional, and we progressively make it smarter."

[PAUSE]

"Our MVP — the minimum viable product — covers phases 1 through 5. Phases 6 and 7 polish and deploy."

[STEP BACK]

Time: ~1:30 minutes (combined for Slides 5 & 6)

Transition to Member 6: "To explain the impact and scalability of our solution, here is [Member 6's name]."

PART 18 — SLIDES 7 AND 8 SCRIPT (MEMBER 6)

"Impact, Scalability & Conclusion"

[STEP FORWARD]

"Thank you, [Member 5]."

Let me explain why CareerFit matters and how it can grow."

EXPECTED IMPACT

[POINT TO SLIDE 7]

"CareerFit directly addresses a critical gap in Indian education — the disconnect between learning and employability."

"First: Better Career Awareness. Students currently have no reliable way to know if they are job-ready. CareerFit gives them a clear, data-driven assessment."

[PAUSE]

"Second: Personalized Learning. Instead of randomly picking courses, students follow a roadmap prioritized by industry demand. This saves time and focuses effort."

[PAUSE]

"Third: Reduced Career Confusion. Many students don't know what skills to learn for their target role. CareerFit answers that question directly."

[PAUSE]

"Fourth: Improved Employability. When students know exactly what to learn and in what order, they become job-ready faster. This benefits students, colleges, and employers."

[PAUSE]

"Fifth: Education-to-Industry Connection. CareerFit creates a feedback loop — students learn what industry actually needs, not just what textbooks teach."

SCALABILITY

[POINT TO SCALABILITY FLOW]

"We start with our college as the initial user base."

From there, it scales to multiple colleges, then universities, then institutions across regions — eventually reaching a larger student ecosystem nationwide."

[PAUSE]

"The platform is designed to be institution-agnostic. Any college can onboard students, and the AI adapts to different career domains — not just software, but also data science, design, management, and more."

FUTURE SCOPE

[LOOK AT JUDGES]

"In future versions, we plan to add:

- Career-path recommendations across multiple job roles
- Skill trend analysis showing which skills are growing or declining in demand
- Institution dashboards for placement coordinators
- Multilingual support for students across India
- An AI career assistant for real-time guidance
- Integration with learning platforms like Coursera, Udemy, and NPTEL
- Industry partnerships for direct placement pathways"

[PAUSE]

"We are not promising everything at once. Our MVP delivers the core value — skill analysis, gap detection, and personalized roadmap. Everything else builds on that foundation."

CONCLUSION

[SLIDE 8 — STEP FORWARD]

"To conclude:

The problem is real — students don't know if they are job-ready. The solution is clear — CareerFit makes the gap visible and actionable. The AI is purposeful — every AI component serves a specific function. The impact is measurable — better career readiness, better employability. The plan is realistic — six students, seven phases, achievable MVP."

[PAUSE — LOOK AT ALL JUDGES]

"We are Team [TEAM NAME]. CareerFit is our answer to the question every student asks:

'Am I ready for the career I want?'

Thank you. We are ready for your questions."

[STEP BACK]

Time: ~1:30 minutes

PART 19 — JUDGE QUESTIONS

30 Questions Judges May Ask

#	Question
1	Why did you choose this problem?
2	Why is this a Smart Education problem?
3	Why does this require AI?
4	Why can't students just use ChatGPT?
5	Where exactly is AI used in your system?
6	What data do you use?
7	Where will the data come from?
8	How do you handle incorrect AI results?
9	How accurate is your system?
10	How do you calculate the readiness score?
11	Why should users trust the score?
12	How is your solution different from LinkedIn or job portals?

13	How is it different from a normal resume checker?
14	What is innovative about your solution?
15	Is the project technically feasible for six students?
16	What is your MVP?
17	What will you build first?
18	How long will development take?
19	How will you scale this?
20	How will you protect user data?
21	What if the AI API fails?
22	What if the resume has unusual formatting?
23	What if a skill has multiple names?
24	How do you handle new or emerging skills?
25	How do you evaluate whether your recommendations are good?
26	How do you prevent AI hallucinations?
27	What are the limitations of your solution?
28	What happens after the internal hackathon?
29	What did each of you personally contribute?
30	Did you use AI to build this project?

PART 20 — ANSWER ABOUT AI USAGE

When Judges Ask: "Did You Use AI to Build This?"

Answer truthfully:

"Yes, we used AI-assisted development tools to accelerate coding, debugging, research, and documentation. However, the team designed the architecture, validated the generated outputs, tested the system, and understands every component we are presenting."

Key Points:

- **Be honest** — Judges respect honesty
- **Be specific** — Mention what AI helped with
- **Emphasize understanding** — You know what was built and why
- **Emphasize validation** — You tested and verified everything
- **Don't apologize** — Using AI tools is standard practice
- **Don't overclaim** — Don't say AI did everything

What Matters to Judges:

- 1. Problem understanding
- 2. Solution design
- 3. Technical architecture
- 4. Team understanding
- 5. Validation and testing
- 6. Real-world impact

The number of AI tools used does not make the project innovative.

PART 21 — WHAT NOT TO DO

15 Things to Avoid

#	Don't Do This
1	Force CareerFit into an unrelated SIH problem statement
2	Start coding before understanding the problem
3	Add AI everywhere just for appearance
4	Put 20 technologies on the slide to look impressive
5	Claim "nobody has ever done this" without evidence
6	Claim impossible accuracy (e.g., "100% accurate")
7	Say "AI will solve everything"
8	Give fake statistics or made-up numbers
9	Copy another team's idea
10	Use plagiarized content
11	Let only one member understand the project
12	Memorize answers without understanding
13	Make the presentation too complicated
14	Spend 5 minutes explaining technology and 30 seconds explaining the problem
15	Present the existing CareerFit as if it was created specifically for SIH

PART 22 — TEAM WORKFLOW

16-Phase Preparation Plan

Phase	Activity	Duration
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1	Understand SIH — rules, format, evaluation criteria	Day 1
2	Study official problem statements carefully	Day 1-2
3	Shortlist 2–3 problems that interest the team	Day 2
4	Research users and existing solutions for each problem	Day 3
5	Choose ONE problem that fits best	Day 3-4
6	Decide whether CareerFit genuinely fits this problem	Day 4
7	Define the new/enhanced CareerFit solution for this problem	Day 5
8	Define MVP — what must be built first	Day 5-6
9	Design AI workflow — where AI is used and why	Day 6
10	Design system architecture	Day 7
11	Choose and justify technology stack	Day 7
12	Create implementation timeline	Day 8
13	Build presentation (8 slides)	Day 8-9
14	Practice six-person presentation (time each speaker)	Day 9-10
15	Practice judge Q&A (every member answers every question)	Day 10-11
16	Stress-test the idea (act as harsh judges)	Day 11-12

PART 23 — STRESS TEST THE IDEA

Before Finalizing — Ask These Questions

#	Question	Weak Answer	Strong Answer
1	Is this actually a real problem?	"We think so"	"Yes, studies show X% of graduates face this"
2	Who specifically has this problem?	"Students"	"Final-year engineering and CS students preparing for campus placements"
3	How big is the problem?	"Very big"	"India produces 1.5M+ graduates annually; industry reports X% are unemployable"
4	Does the solution actually solve it?	"Yes"	"It directly addresses the gap by doing X, Y, Z"
5	Why does AI help?	"Because AI is smart"	"Because skill matching requires semantic understanding, not keyword matching"
6	Why can't a rule-based system solve it?	"It can"	"It partially can, but cannot handle synonyms, context, or personalized roadmaps"

7	What is genuinely innovative?	"We use AI"	"Explainable readiness scoring with prioritized learning paths — not just matching"
8	What existing solutions exist?	"None"	"LinkedIn, resume checkers, course platforms exist, but none combine all three aspects"
9	What makes ours different?	"We are better"	"We provide explainable scoring + prioritized roadmap + continuous tracking — in one platform"
10	Can six students build this?	"Yes"	"Yes, with a clear MVP, phased approach, and existing CareerFit foundation"
11	What is the MVP?	"Everything"	"Resume parsing + skill extraction + matching + scoring + roadmap — core 10 features"
12	What data do we need?	"Resumes"	"User-uploaded resumes + job descriptions + skill database + course database"
13	Can we legally use the data?	"Yes"	"Yes — users voluntarily upload their own data; we don't scrape or buy data"
14	What if AI is wrong?	"It won't be"	"We have validation layers, fallback mechanisms, and user feedback loops"
15	How will we measure success?	"Users like it"	"Accuracy of skill extraction, user satisfaction scores, readiness improvement over time"
16	Can it scale?	"Yes"	"Modular architecture, cloud deployment, multi-college onboarding supported"
17	Is the architecture realistic?	"Yes"	"Each component uses proven technologies; no experimental or untested dependencies"
18	Are we overengineering?	"No"	"MVP is focused; advanced features are in V2/V3"
19	Are we making unsupported claims?	"No"	"Every claim backed by research, existing data, or clear reasoning"
20	Can every team member explain the project?	"Mostly"	"Yes — each member trained on all topics, even though they specialize in one"

If any answer is weak, fix it before the presentation.

PART 24 — CAREERFIT MVP

What We Build First (Must Have)

#	Feature
1	Student profile creation
2	Resume upload (PDF/DOCX)

3	Target job selection/input
4	Resume skill extraction (AI)
5	Job requirement extraction (AI)
6	Skill matching (semantic)
7	Missing skill detection
8	Weighted skill priority scoring
9	Career readiness score (explainable)
10	Personalized learning roadmap

What Comes Later (V2)

#	Feature
11	AI career assistant
12	Progress tracking
13	Course recommendations
14	Skill trend analysis
15	Institution dashboard
16	Placement analytics
17	Multilingual support
18	Advanced recommendations
19	Industry partnerships

PART 25 — PROJECT DEMO

Future Demo Flow (After Prototype)

Step	Student Action	System Response
1	Opens CareerFit	Sees welcome screen
2	Creates profile	Profile saved
3	Uploads resume	Resume parsed
4	Selects target job	Job requirements extracted
5	Clicks "Analyze"	AI processing begins

6	Waits briefly	Skills extracted and matched
7	Views results	Matched skills, missing skills shown
8	Scrolls down	Readiness score with explanation
9	Scrolls further	Priority-ordered skill gaps
10	Views roadmap	Personalized learning path displayed

The demo tells a story: A student goes from "I don't know where I stand" to "Now I know exactly what to learn."

PART 26 — PRESENTATION DESIGN

Slide Design Rules

Rule	Detail
One clear title	Each slide has ONE topic
3–6 bullet points max	No paragraphs
Large readable text	Minimum 24pt font
Consistent design	Same colors, fonts, layout
Minimal animations	No distracting transitions
Diagrams where helpful	User journey, AI workflow, architecture

Key Diagrams to Include

#	Diagram	Slide
1	Problem illustration	Slide 1
2	User journey flow	Slide 2
3	AI workflow pipeline	Slide 3
4	System architecture	Slide 4
5	Implementation timeline	Slide 6

PART 27 — FINAL PRESENTATION STORY

The Narrative Arc

MEMBER 1: "Students face a career-readiness problem."

↓

MEMBER 2: "We propose CareerFit to solve it."

↓

MEMBER 3: "AI analyzes the student's skills and identifies the gap."

↓

MEMBER 4: "Here is how the technical system works."

↓

MEMBER 5: "Here is how we will build it."

↓

MEMBER 6: "Here is the impact and how it can scale."

↓

"Thank you. We are ready for your questions."

The entire presentation should feel like ONE continuous story — not six separate speeches.

PART 28 — FINAL OUTPUT REQUIRED

Complete SIH Preparation Checklist

#	Item	Status
1	Final project title	CareerFit: AI-Powered Career Readiness & Skill Gap Platform
2	One-line project pitch	An AI-powered platform that tells students how job-ready they are, why, and what to learn next
3	Problem statement	PROBLEM STATEMENT NOT YET SELECTED
4	Problem justification	Pending problem statement selection
5	Target users	Final-year college students preparing for placements/careers
6	User pain points	Don't know skill gaps, don't know what to learn, no personalized guidance
7	Existing solution limitations	LinkedIn doesn't analyze readiness; resume checkers don't provide learning paths
8	Proposed solution	CareerFit — AI-powered skill analysis + gap detection + personalized roadmap
9	Unique value proposition	Explainable readiness scoring with prioritized learning paths
10	Innovation	Combines skill extraction, semantic matching, weighted scoring, and roadmap generation
11	AI workflow	Document parsing → NLP extraction → normalization → matching → scoring → recommendations
12	AI technologies	NLP, Embeddings, Semantic Similarity, LLM (explanations), ML (scoring)
13	Data flow	User input → Backend → AI Engine → Results → Frontend → User

14	System architecture	■ Frontend (React) → API (FastAPI) → Backend → DB + AI Engine → Results
15	Database design	■ Students, Resumes, Skills, Jobs, Analyses, Learning Paths
16	API design	■ Auth endpoints, Resume endpoints, Analysis endpoints, Profile endpoints
17	Technology stack	■ React/Next.js, Python/FastAPI, PostgreSQL/Supabase, NLP/Embeddings/LLM
18	MVP	■ Resume upload + skill extraction + matching + scoring + roadmap
19	Future features	■ AI assistant, progress tracking, institution dashboard, multilingual
20	Implementation timeline	■ 7 phases from research to deployment
21	Testing strategy	■ Unit tests, integration tests, accuracy validation, user testing
22	AI evaluation strategy	■ Precision/recall for extraction, user feedback for recommendations
23	Security/privacy approach	■ JWT auth, user-owned data, no scraping, encrypted storage
24	Expected impact	■ Better career awareness, personalized learning, improved employability
25	Scalability	■ College → multi-college → university → national student ecosystem
26	Risks	■ AI inaccuracy, resume parsing failures, data privacy concerns
27	Limitations	■ Dependent on resume quality, AI not perfect, limited to uploaded data
28	Mitigation strategies	■ Fallback mechanisms, validation layers, user feedback, manual input option
29	Eight-slide presentation	■ Created (Slides 1–8)
30	Six-member speaking scripts	■ Created (Parts 13–18)
31	Time allocation	■ Created (Part 11)
32	Transition sentences	■ Included in each script
33	Judge Q&A	■ 30 questions prepared (Part 19)
34	Difficult judge questions	■ Covered in stress test (Part 23)
35	Strong answers	■ Provided throughout
36	Demo plan	■ 10-step demo flow (Part 25)
37	Team responsibilities	■ 6 roles defined (Part 9)
38	Individual responsibilities	■ Detailed per role (Part 9)
39	Team practice plan	■ 16-phase workflow (Part 22)

IMPORTANT NOTE

PROBLEM STATEMENT NOT YET SELECTED

This guide is prepared **without an official SIH problem statement** because none has been provided.

What this means:

- The presentation scripts reference a general career-readiness problem
- Once you select the actual SIH problem statement, **adapt CareerFit to it**
- Do NOT present an invented problem as the official SIH problem statement
- If CareerFit doesn't fit the selected problem, the team should explore other solutions

Next Steps:

1. Study the official SIH problem statements when released
2. Shortlist 2–3 problems
3. Research users and existing solutions
4. Select one problem
5. Adapt CareerFit specifically to that problem
6. Update all presentation materials accordingly

APPENDIX — QUICK REFERENCE CARDS

For Every Team Member — 11 Things to Know

#	Topic	One-Line Answer
1	Problem	Students don't know if they are job-ready
2	Users	Final-year college students, job seekers
3	Solution	AI-powered skill analysis + gap detection + learning roadmap
4	AI	NLP extracts skills, embeddings match semantically, LLM generates roadmaps
5	Architecture	Frontend → API → Backend → AI Engine → Database → Results
6	Innovation	Explainable readiness scoring with prioritized learning paths
7	Stack	React/Next.js, Python/FastAPI, PostgreSQL, NLP/Embeddings/LLM
8	MVP	Resume upload + skill extraction + matching + scoring + roadmap
9	Impact	Better career readiness, personalized learning, improved employability
10	Limitations	AI accuracy depends on resume quality; not perfect
11	Future	AI assistant, institution dashboard, multilingual, industry partnerships

